Ledergor 2018 Multiple Myeloma — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=15) fuses the patient-private myeloma clones: same-sample neighbour fraction 0.87→0.26, iLISI 1.0→5.1 / 16, one connected kNN component. Judged on the regenerated embedding across a θ-sweep — samples intermix on one continuous, uniformly low-cycling body (no proliferative lobe to scope).

Gate 2 — module geography → GO (non-branching)

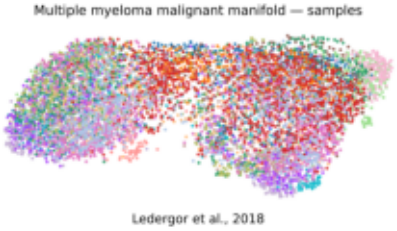
Signatures collapse onto a single (Gast/pSom)→Crest axis: smoothed pSom↔Crest r=−0.97, Gast↔pSom r≈−0.10 (no cross-repression → no two competing arms). Crest is an inverse terminal gradient (crest-high decile pseudotime 0.83; joint pSom&Crest enrichment 0.16). Principal curve is fold-free (2 tips / 0 forks).

Gate 3 — hub cross-repression → NA

Non-branching topology — no bifurcation and no Gast/pSom hub, so the branching-only cross-repression test does not apply. Correlation & modules notebooks are correctly not produced.

STEP FIGURES

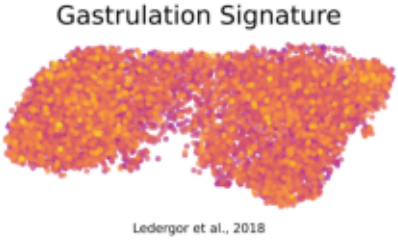
Manifold — samples



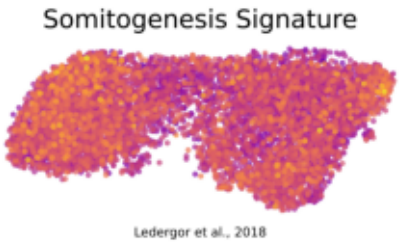
Pseudotime



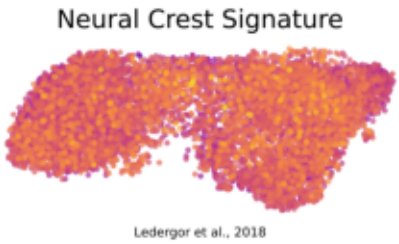
Gastrulation



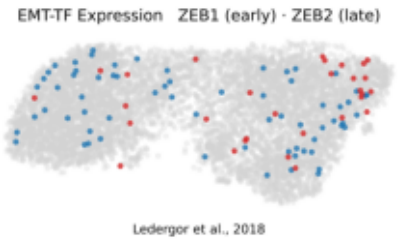
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Multiple myeloma (Ledergor et al., 2018), MARS-seq — 8,182 malignant plasma cells over 16 samples. Uniformly low-cycling body (no proliferative lobe to scope).

Verdict.
Pattern reproduces as the NON-BRANCHING case — a single gastrulation/somitogenesis→neural-crest developmental axis; no hub, no bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB1 (root) → ZEB2 (terminus) — the only detectably-expressed factors.

Parameter choices

Min-cell threshold	≥95 (geom-mean of 183→50 cliff; 16/18 samples; drops MM04, MM05)	**Integration**	Harmony on sample, θ=15 (patient-private MM clones), n_pcs=40, k=30, spread=1.3
Islands removed	15 cells / 2 detached specks (UMAP radius graph, largest component)	**Geometry**	Non-branching → scFates curve (ElPiGraph) N=20 λ=0.3 μ=100; 2 tips / 0 forks, fold-free
Root node	curve tip 0 — Gast/pSom-high, Crest-low (pSom→Crest along pseudotime)	**Module means (raw)**	Gast −0.029 · pSom +0.016 · Crest +0.004 → Linear-pSom/Crest
EMT-TFs shown	ZEB1 (root) → ZEB2 (terminus); detectable 3.3% / 1.5%		